Enjoyable Physical Therapy Experience with Interactive Drawing Games in Immersive Virtual Reality

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Overview

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Introduction: Physical Therapy (PT)

- Advantages of Conventional PT:
  - One-on-one use of medical resources
  - Effective in improving mobility
  - Avoid surgery

- Disadvantages of Conventional PT:
  - Lack of motivation
  - Slow recovery progress
  - Absence of support

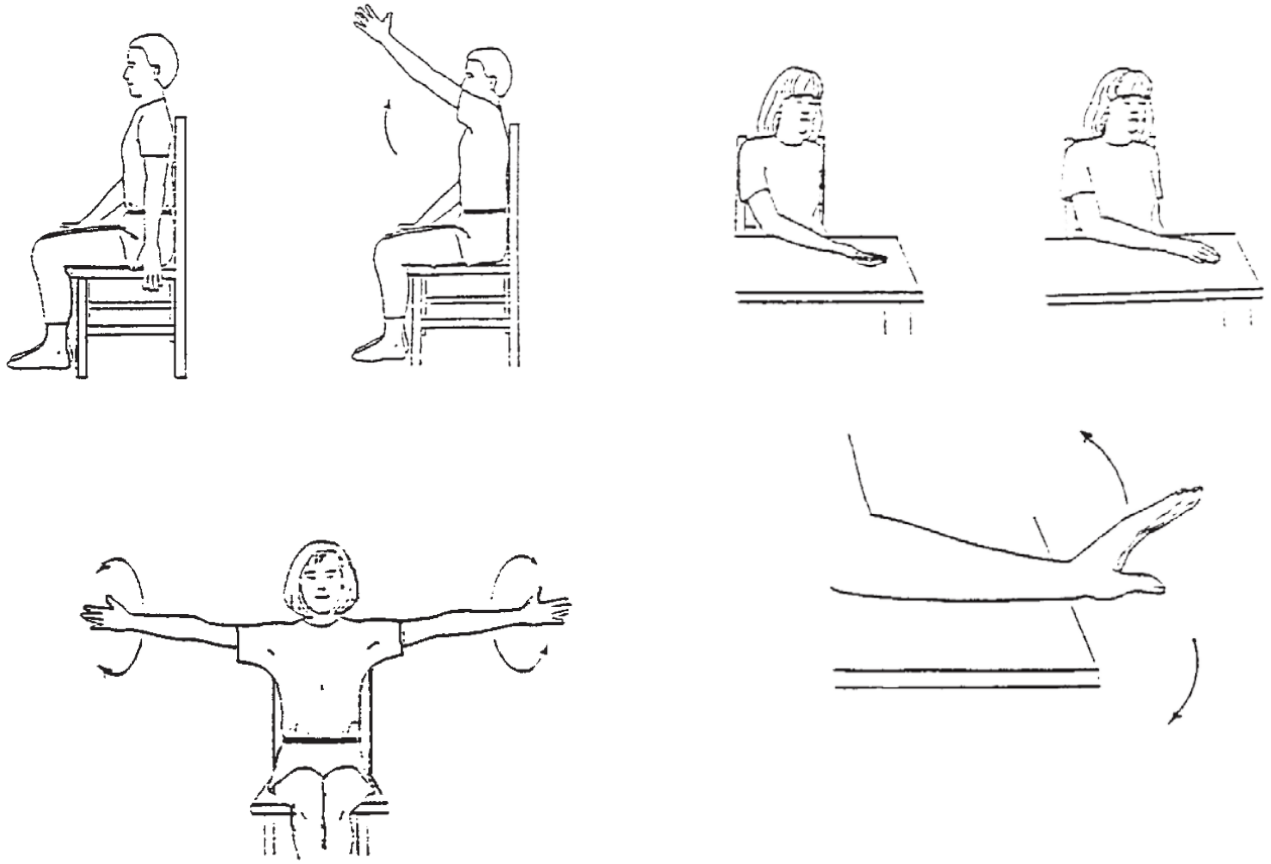


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Introduction: Virtual Reality (VR) Therapy

- Potential for VR in PT
  - Engagement with appealing visual effects, interactive sensory feedback
  - Portability for home-therapy
  - Ability to attenuate pain
  - Independence from external pressures/distractions
  - Controlled environment to easily measure and evaluate performance



Photo Credit: https://www.saebo.com/shop/saebovr/

- Hesitations of using VR in PT
  - Customization of therapy intensity
  - Influences on the user's therapy experience

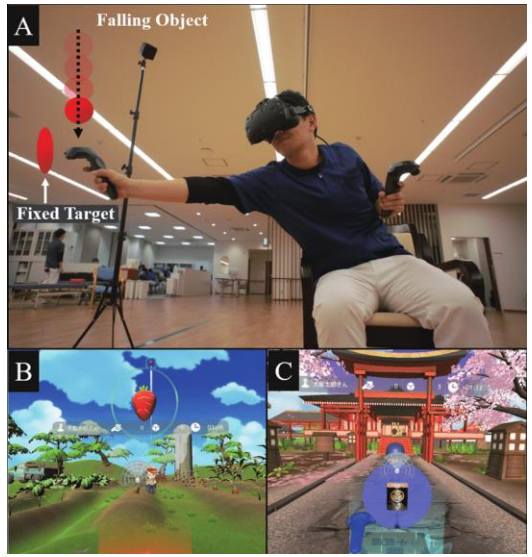


Photo Credit: doi.org/10.2490/prm.20190011

Related Works

VRT-Home, Li et al.

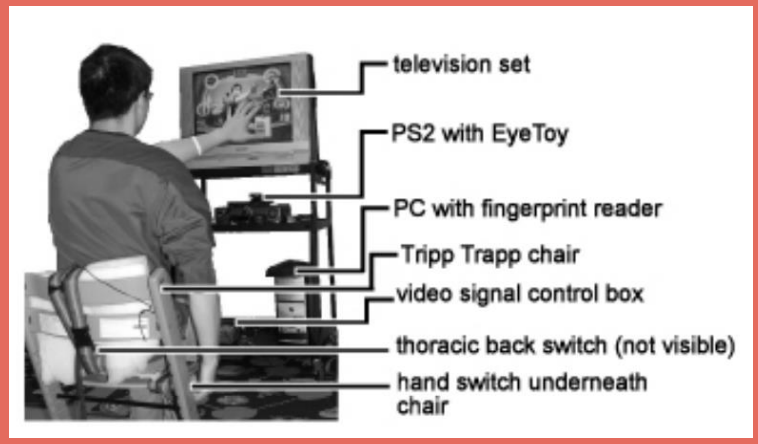


Photo Credit: doi.org/10.3233/TAD-2009-0277

iVR exergame, Xu et al.

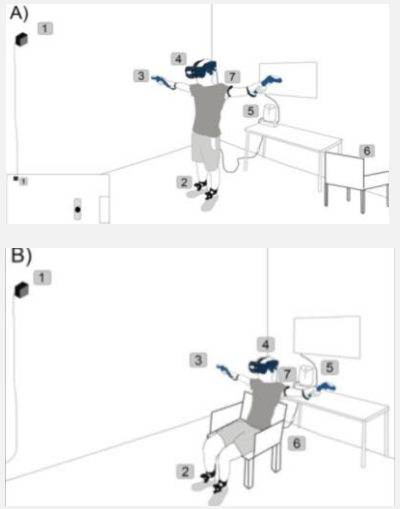


Photo Credit: doi.org/10.2196/17972

Jabberstamp, Raffle et al.



Photo Credit: doi.org/10.1145/1297277.1297306

Tilt Brush, Google VR



Photo Credit: https://www.youtube.com/watch?v=TckqNdrdbgk

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Research Questions

RQ1: Do simple adjustments of VR contents **influence the subjective** perception of difficulty/**intensity**?

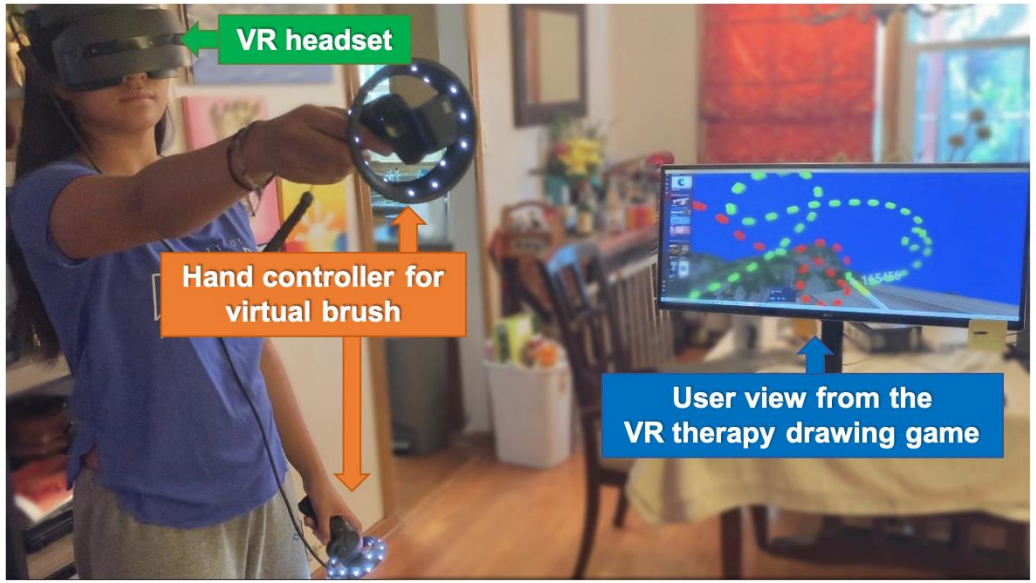
RQ2: How does the user's position influence the VR **drawing therapy** experience?

RQ3: How is the enjoyment of the drawing **game associated** with the difficulty of the therapy?

Experiment: VR Drawing Game

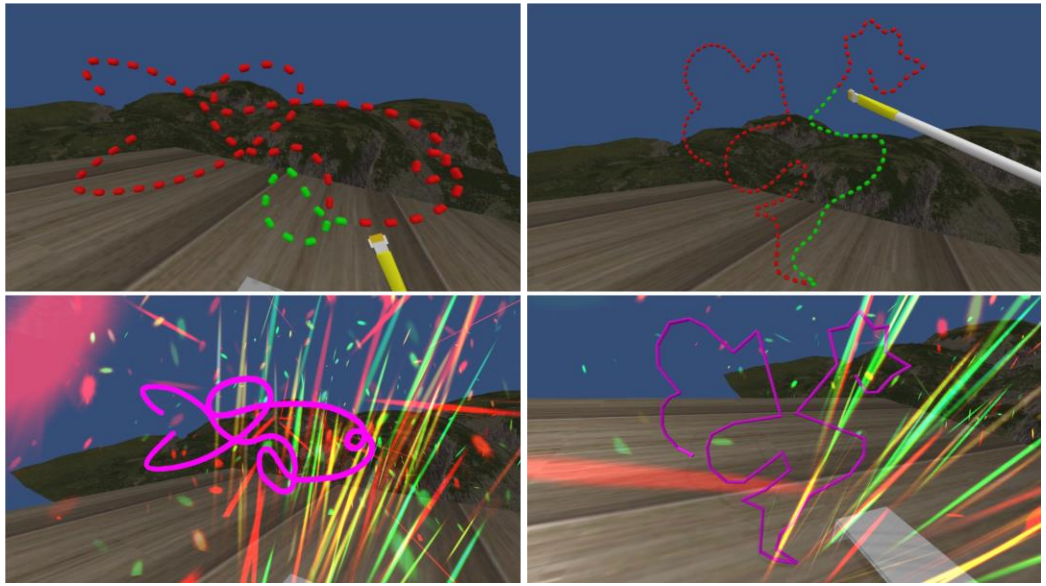
- Apparatus- creative line art drawing game in an immersive VR environment as a tool for enjoyable upper extremity physical therapy

- Procedure-
  - Assign participants
  - Pre-questionnaire
  - Conduct tests
  - Post-questionnaire



Experiment: Study Design

- Study Design (2x2 mixed design)
  - Between subjects – Content – Easy (Fish) vs Hard (Chicken)
  - Within subjects – Position – Seated vs Standing
- Participants
  - 16 non-clinical volunteers enrolled in the University of Delaware's undergraduate program
- Measures
  - Objective – Task Completion Time, Normalized Number of Mistakes
  - Subjective – Easiness, Comfort, Body Stretch, Enjoyment, Willingness to Recommend



(a) Easy (Fish) (b) Hard (Chicken)

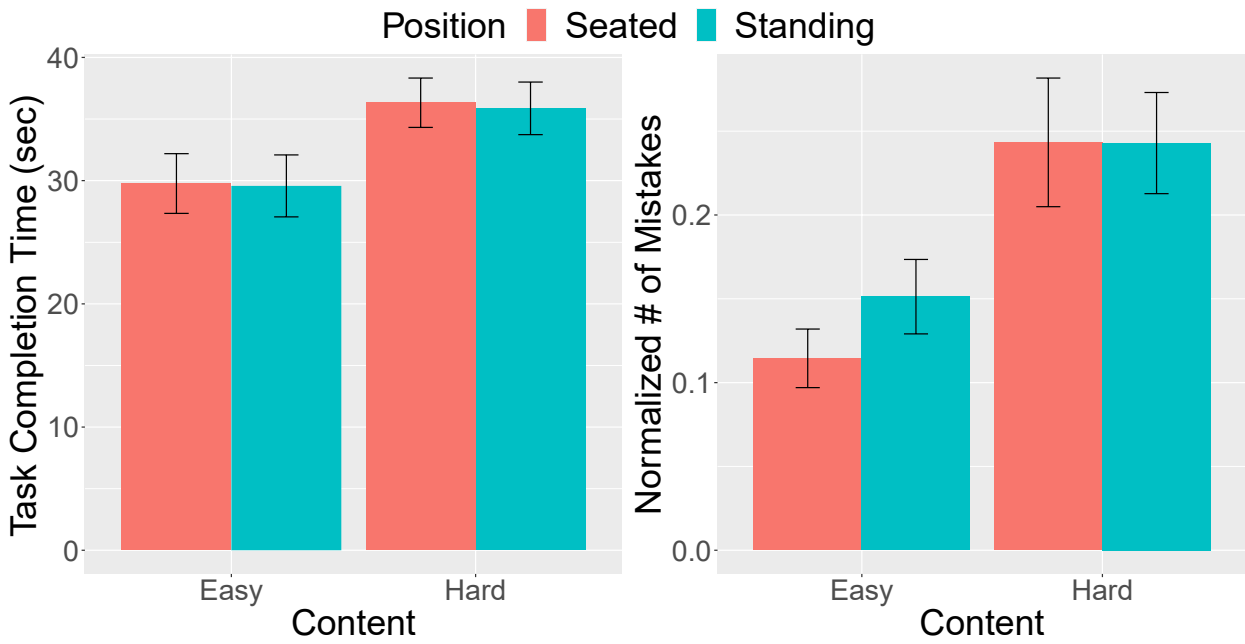
Hypotheses

- H1: Performance and perception will be influenced by content
  - H1.1: Performance will be reduced with the Hard content
  - H1.2: Perception of easiness and comfort will be improved with the Easy content
- H2: Therapy experience will be influenced by position
  - H2.1: Participants will perceive more body stretch in the Standing position
  - H2.2: Participants will enjoy the exercise more in the Standing position



Results: Objective Measures

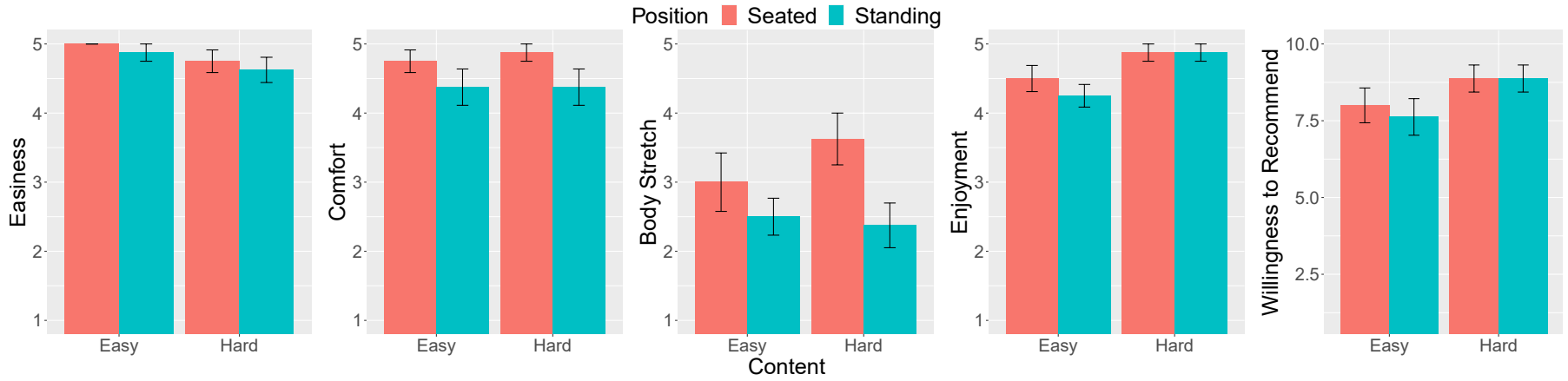
- Normalized Number of Mistakes: main effect of the Content factor (p-value < 0.05)
- Task Completion Time: no observed effect



Results: Subjective Measures

- Body Stretch: main effect of the Position factor (p-value < 0.05)
- Enjoyment: main effect of the Content factor (p-value < 0.05)
- Easiness, Comfort, Willingness to Recommend: no observed effect

Participant #4: *"When I was sitting I had to reach more and wanted to lift myself off the chair a bit to get to the highest parts of the chicken. But since I couldn't I had to really stretch my arm and controller out to hit the dots."*



Discussion

- **Content Factor:** influences objective measures,

but no evidence of influence on subjective measures

- **Position Factor:** influences body stretch,

but no evidence of influence on enjoyment (both subjective measures)

Conclusion

We presented a creative VR therapy game for upper extremity rehabilitation using line drawing activities

We investigated how the drawing content and the user's position could influence the user's performance and perception

Our results showed that our VR drawing game is an effective tool for enjoyable physical therapy, which has a lot of potential for a customized therapy experience

We will further investigate the broader use cases of our interactive VR drawing intervention, i.e., other physical therapy exercises



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Q & A







